

MAT 310, Handout 14: Normal (Gaussian) Distribution (4.5)

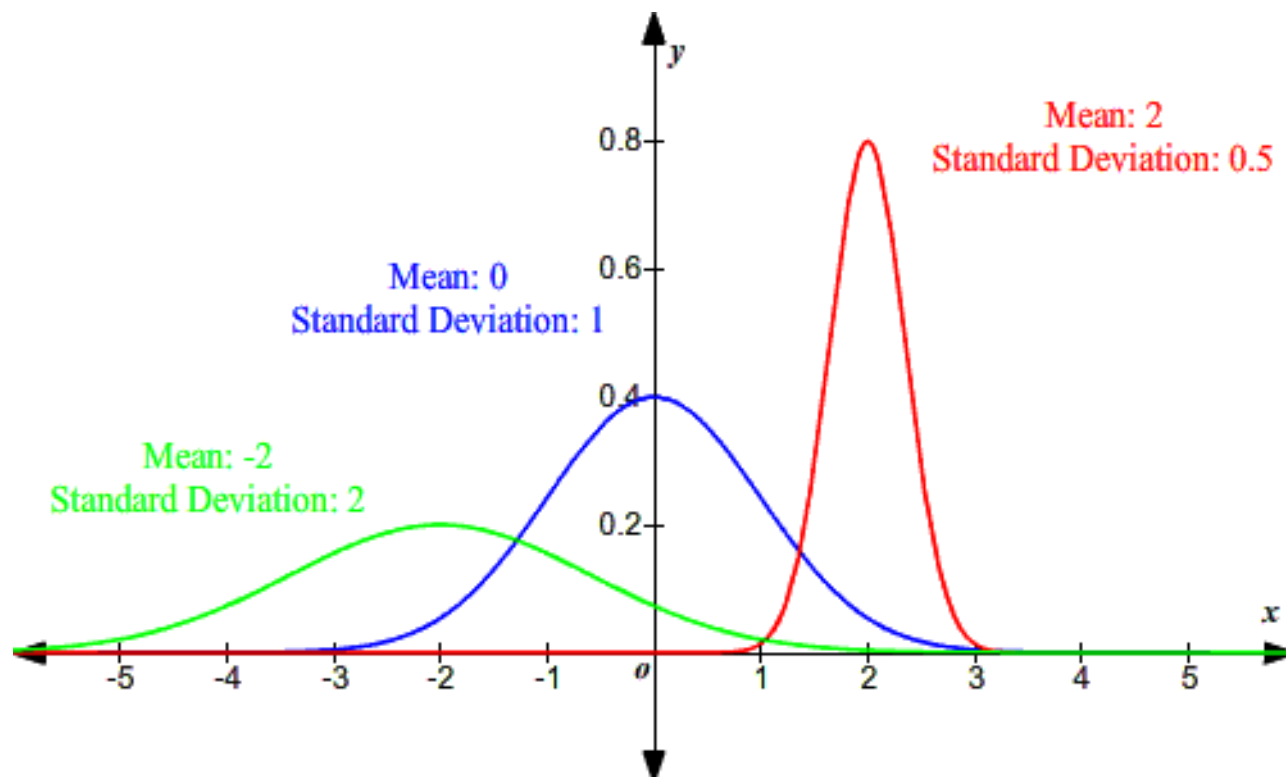
Normal Distribution $Y \sim N(\mu, \sigma)$ is normally distributed with mean μ and standard deviation σ if its density function is given by:

$$f(y) =$$

Remarks:

- The Normal distribution is ubiquitous in statistics, primarily due to the Central Limit Theorem.
- There is no closed form of $F(y)$ for $Y \sim N(\mu, \sigma)$.
- Despite our inability to integrate $f(y)$, we do know that $\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy =$
- We refer to μ as the _____ parameter and σ as the _____ parameter.

Illustration:



Standard Normal Distribution: We refer to $Z \sim N(0, 1)$ as the standard normal RV. Because Z is symmetric about 0, we know that, for positive z :

$$P(Z > z) = P(Z < -z).$$

Note: standard normal probabilities can be found by using either statistical tables or statistical software.

TI-83/84 calculator:

Example 1. Use a calculator to find the following:

- (a) $P(Z < 0)$
- (b) $P(Z > 1.15)$
- (c) $P(Z < 2.10)$
- (d) $P(-1 < Z < 1.42)$
- (e) $P(-.68 < Z < -.35)$

Important Remark: The $(100 \cdot p)^{th}$ percentile of Z , z_0 , can be found by using either statistical tables or statistical software.

TI-83/84 calculator:

Example 2. Find the value z_0 such that:

- (a) $P(Z > z_0) = .2$
- (b) $P(Z < z_0) = .9$
- (c) $P(-z_0 < Z < z_0) = .95$

Linear Functions of Normal RVs are also Normal. Consider $X \sim N(\mu, \sigma)$. We also know that:

- $X + b \sim$
- $aX + b \sim$

Question: How do we find probabilities for non-standard Normal RVs? One method is _____.
Given $X \sim N(\mu, \sigma)$, the standardized value is given by

$Z =$

Note: Z gives the number of standard deviations that X is either above or below μ .

Example 3. Given $X \sim N(5, 2)$, find:

- (a) $P(X < 3)$
- (b) the 90th percentile of X

Remark: Alternatively, you can type the mean and standard deviation directly into the calculator/software.